Cybersecurity Lab Setup Report

ApexPlanet Internship – Task 1: Foundation & Environment Setup

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1. Introduction

Cybersecurity protects computers, networks, applications and data from unauthorized access, misuse, disruption and damage. A controlled virtual lab provides a safe environment for learning system administration and network analysis.

2. Objectives

• Understand cybersecurity fundamentals and the CIA Triad.

• Identify common cyber threats and attack vectors.

• Configure an isolated VirtualBox lab.

• Set up Kali Linux and Metasploitable2.

• Verify network connectivity.

• Perform authorized service discovery with Nmap.

• Capture and analyze ICMP packets with Wireshark.

3. CIA Triad

• Confidentiality: Information is accessible only to authorized users.

• Integrity: Information remains accurate and is not changed without authorization.

• Availability: Systems and data are accessible when needed.

4. Cyber threats

Malware includes viruses, worms, Trojans, ransomware and spyware. Other common threats include phishing, password attacks, denial of service, man-in-the-middle attacks, insider threats and social engineering.

5. Attack vectors

Common paths include exposed network services, weak or reused passwords, phishing links and attachments, unpatched software, misconfiguration, removable media, third-party dependencies and physical access.

6. Requirements

Hardware: A computer with sufficient memory, storage and processor capacity to run the virtual machines. Add your actual laptop specifications here.

Software: Oracle VirtualBox, Kali Linux, Metasploitable2, Nmap and Wireshark.

7. VirtualBox setup

Install VirtualBox from the official Oracle website, open VirtualBox Manager, import or create the required virtual machines, and verify their memory, processor and storage settings before starting them.

8. Kali Linux setup

Start Kali Linux, sign in using the configured account, open a terminal and check the interface address:

ip addr

The lab screenshot showed `eth0` with `192.168.56.103/24`.

9. Metasploitable2 setup

Import the Metasploitable2 VM. In VirtualBox Settings → Network, enable Adapter 1 and set Attached to to Host-only Adapter. Select the VirtualBox Host-Only Ethernet Adapter. Start the VM and use the lab-provided login. Keep this deliberately vulnerable VM isolated.

10. Host-Only network

Host-Only networking lets the host and selected VMs communicate through a private virtual network. Addresses observed in the lab were Kali `192.168.56.103` and Metasploitable2 `192.168.56.101`. Confirm current addresses with `ip addr` and `ifconfig`.

11. Linux fundamentals

Linux is an operating system used on servers, workstations and security tools. Common commands include `pwd` (current directory), `ls -la` (list files), `cd` (change directory), `whoami` (current user), `ip addr` (network addresses) and `ps aux` (process list).

12. Networking fundamentals

An IP address identifies a network interface. A subnet prefix defines the network portion. A MAC address is used at the link layer. TCP is connection-oriented; UDP is connectionless; ICMP supports diagnostics; DNS resolves names; DHCP assigns network configuration; and ports identify application endpoints.

13. Connectivity test

From Kali, the lab used:

ping -c 4 192.168.56.101

An initial attempt returned `Destination Host Unreachable`. After correcting the network setup, a later attempt received four replies with 0% packet loss. This confirms connectivity at the time of the successful test.

14. Nmap service discovery

Nmap is used for network discovery and security auditing. The lab command was:

nmap -sV 192.168.56.101

The screenshot showed one host up and multiple open TCP services, including web, file-sharing, remote-login, FTP, database, VNC and Tomcat-related services. Exact results may vary.

15. Wireshark packet analysis

Wireshark captures and inspects network traffic. Select `eth0`, start capture, generate traffic using the ping command, stop capture and apply the display filter `icmp`. Inspect the IPv4 and ICMP sections, then save the capture as `.pcapng`.

Packet 5 – Echo Request

The supplied screenshot shows source `192.168.56.103`, destination `192.168.56.101`, ICMP type 8 (Echo Request), code 0 and sequence number 1. Wireshark identifies frame 6 as the response. This is Kali asking the target to reply.

Packet 6 – Echo Reply

The supplied screenshot shows source `192.168.56.101`, destination `192.168.56.103`, ICMP type 0 (Echo Reply), code 0, request frame 5 and response time about 1.006 ms. This is the target replying to Kali.

Field	Packet 5	Packet 6
Direction	Kali → Target	Target → Kali
Source	192.168.56.103	192.168.56.101
Destination	192.168.56.101	192.168.56.103
ICMP type	8 – Echo Request	0 – Echo Reply

16. Cryptography and OpenSSL

Encryption converts plaintext into ciphertext; decryption restores it using the required key. Symmetric cryptography uses one shared secret key. Asymmetric cryptography uses a public/private key pair. Hashing produces a fixed-length digest and is not reversible encryption. Digital signatures help verify authenticity and integrity. OpenSSL is a toolkit for cryptographic operations and certificate handling. Check its availability with `openssl version`. Only claim practical completion if you performed it and saved evidence.

17. Results and learning outcomes

The lab used Kali Linux and Metasploitable2 on a Host-Only network. Connectivity was verified using ping. Nmap was used to inspect target services, and Wireshark captured and displayed the ICMP Echo Request and Echo Reply.

18. Conclusion

Task 1 provided practical experience with virtual lab setup, network isolation, Linux commands, connectivity testing, service discovery and packet-level analysis. Continue all security testing only in authorized environments.

19. Evidence to attach

Add genuine screenshots of VirtualBox, both VMs, Host-Only settings, IP configuration, successful ping, Nmap output, Wireshark packet list, Packet 5 and Packet 6. Include the capture file only if the assignment permits it.

20. References

• Kali documentation: https://www.kali.org/docs/

• VirtualBox manual: https://www.virtualbox.org/manual/

• Nmap reference: https://nmap.org/book/man.html

• Wireshark documentation: https://www.wireshark.org/docs/

• OpenSSL documentation: https://docs.openssl.org/